



NEW WHEELS PROJECT REPORT

SQL



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PGP DATA SCIENCE

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Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers. New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

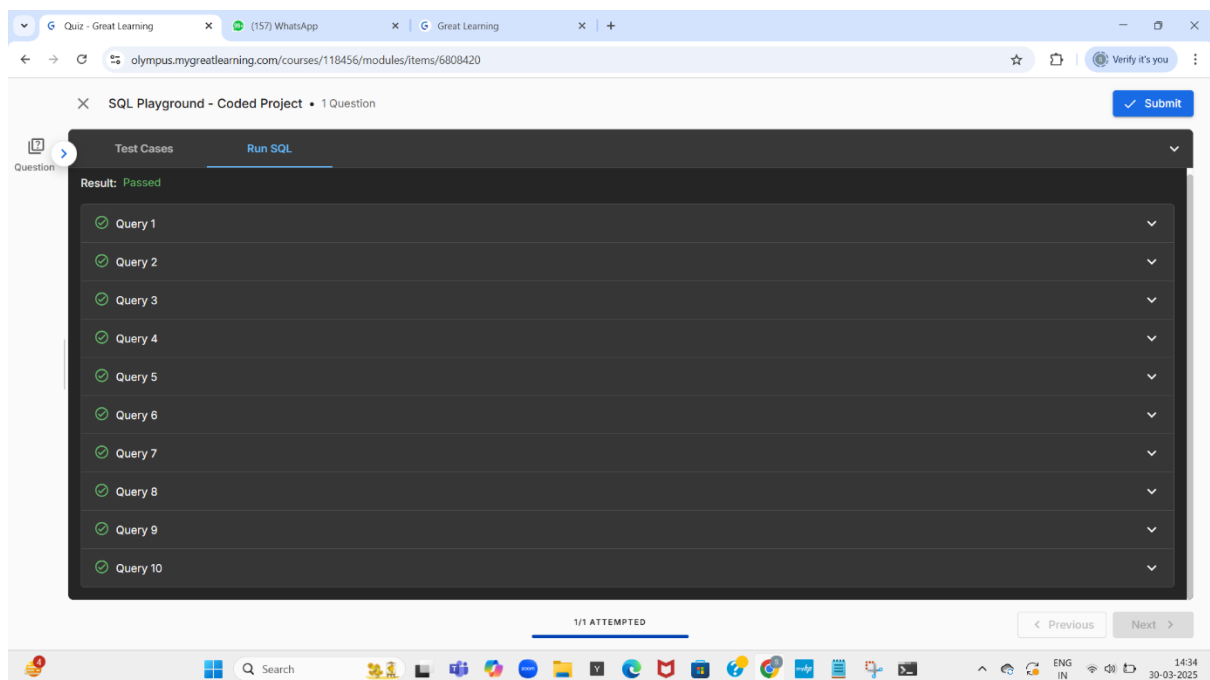
Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Business Questions

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states? [4 marks]



SOLUTION QUERY:

```
SELECT
    state,
    COUNT(customer_id) AS Num_Customers
FROM customer_t
GROUP BY state
ORDER BY 2 DESC;
```

OUTPUT:

Quiz - Great Learning (156) WhatsApp Great Learning olympus.mygreatlearning.com/courses/118456/modules/items/6808420

SQL Playground - Coded Project • 1 Question

Test Cases Run SQL

Result: **Passed**

Query 1

Query:

```
SELECT
  state,
  COUNT(customer_id) AS Num_Customers
FROM customer_1
GROUP BY state
ORDER BY 2 DESC
```

Output:

Showing first 10 rows out of 49 rows

state	Num_Customers
Texas	97
California	97
Florida	86

1/1 ATTEMPTED

Previous Next

14:15 30-03-2025

Quiz - Great Learning (156) WhatsApp Great Learning olympus.mygreatlearning.com/courses/118456/modules/items/6808420

SQL Playground - Coded Project • 1 Question

Test Cases Run SQL

state	Num_Customers
Texas	97
California	97
Florida	86
New York	69
District of Columbia	35
Ohio	33
Colorado	33
Alabama	29
Washington	28
Arizona	26

1/1 ATTEMPTED

Previous Next

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Observations and Insights:

The highest number of customers are from **California and Texas (97 each)**, followed by **Florida (86)** and **New York (69)**.

The lowest number of customers are from **Wyoming, Vermont, and Maine (1 each)**.

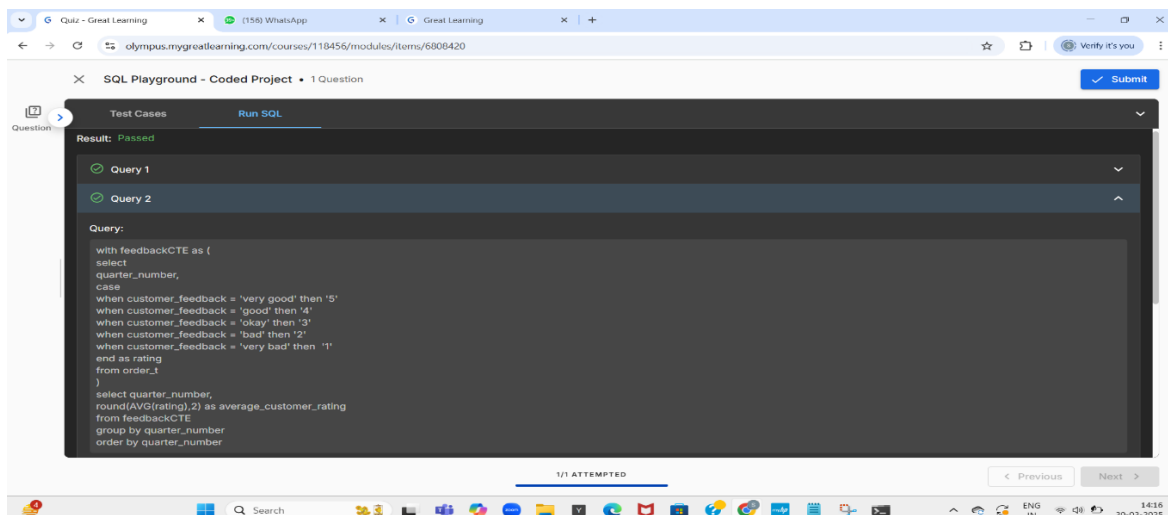
The concentration of customers is significantly higher in large, populous states, while smaller states have fewer customers.

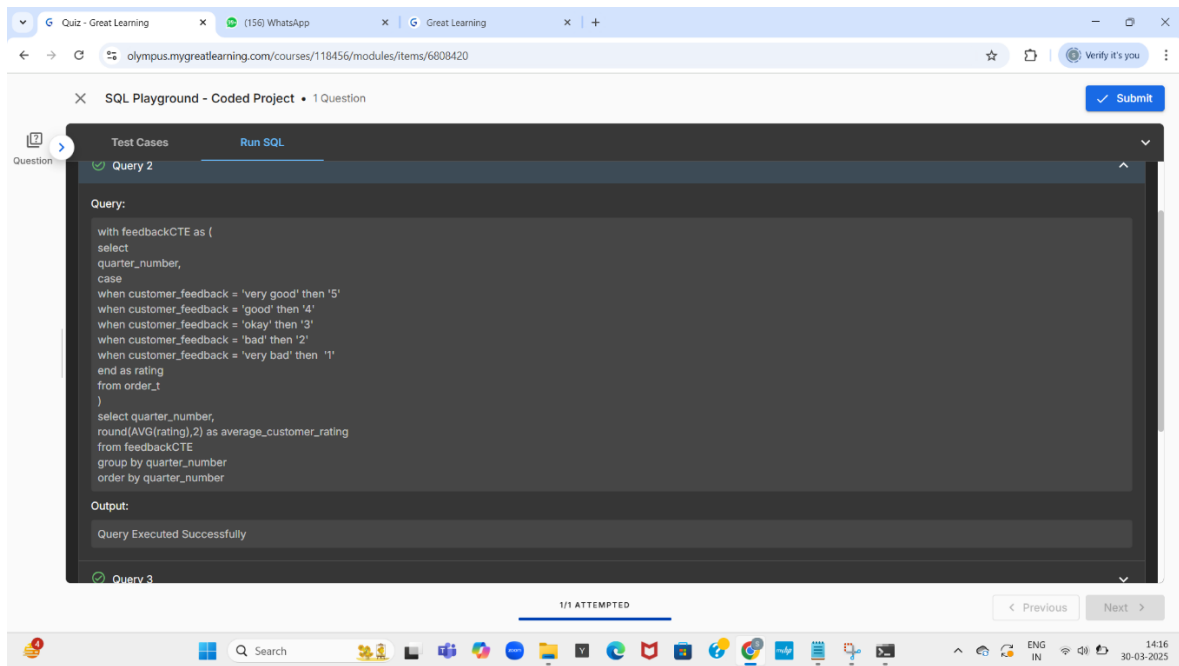
Question 2: Which are the top 5 vehicle makers preferred by the customers? [4 marks]

SOLUTION QUERY:

```
with feedbackCTE as (  
select  
quarter_number,  
case  
when customer_feedback = 'very good' then '5'  
when customer_feedback = 'good' then '4'  
when customer_feedback = 'okay' then '3'  
when customer_feedback = 'bad' then '2'  
when customer_feedback = 'very bad' then '1'  
end as rating  
from order_t  
)  
select quarter_number,  
round(AVG(rating),2) as average_customer_rating  
from feedbackCTE  
group by quarter_number  
order by quarter_number;
```

OUTPUT:





Observations and Insights:

Customer ratings are declining over time:

- **Q1: 3.55** (Closer to "Good")
- **Q2: 3.35** (Decline towards "Okay")
- **Q3: 2.95** (Closer to "Okay")
- **Q4: 2.40** (Between "Bad" and "Okay")

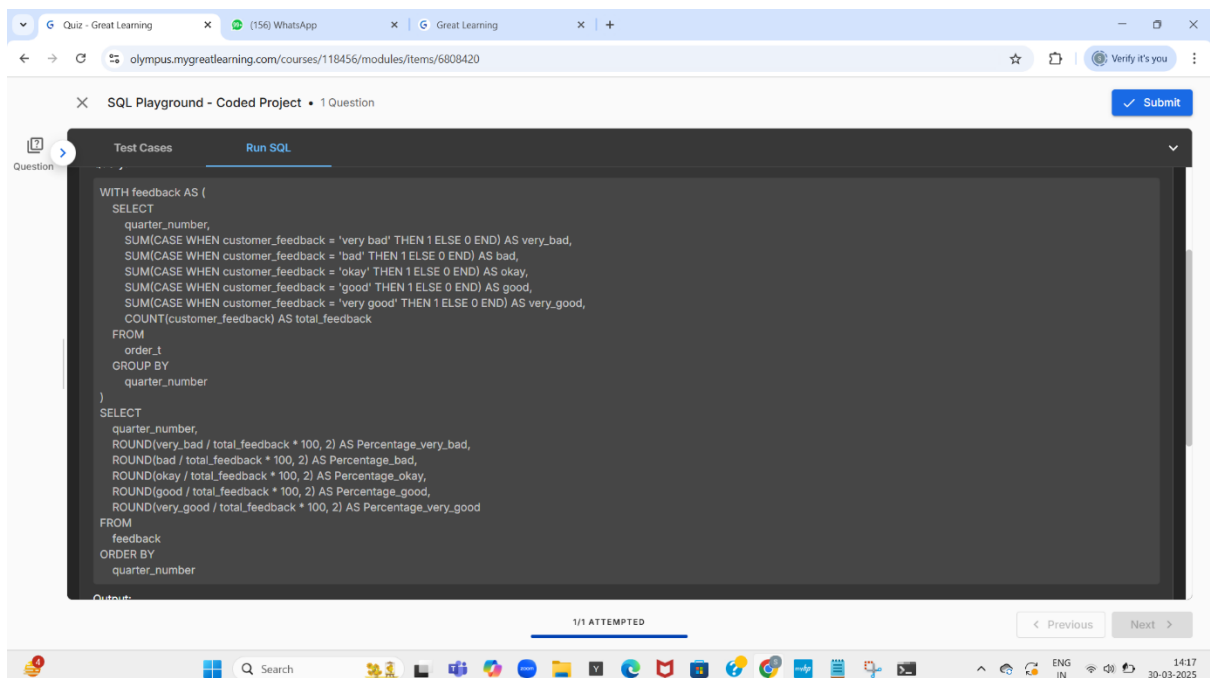
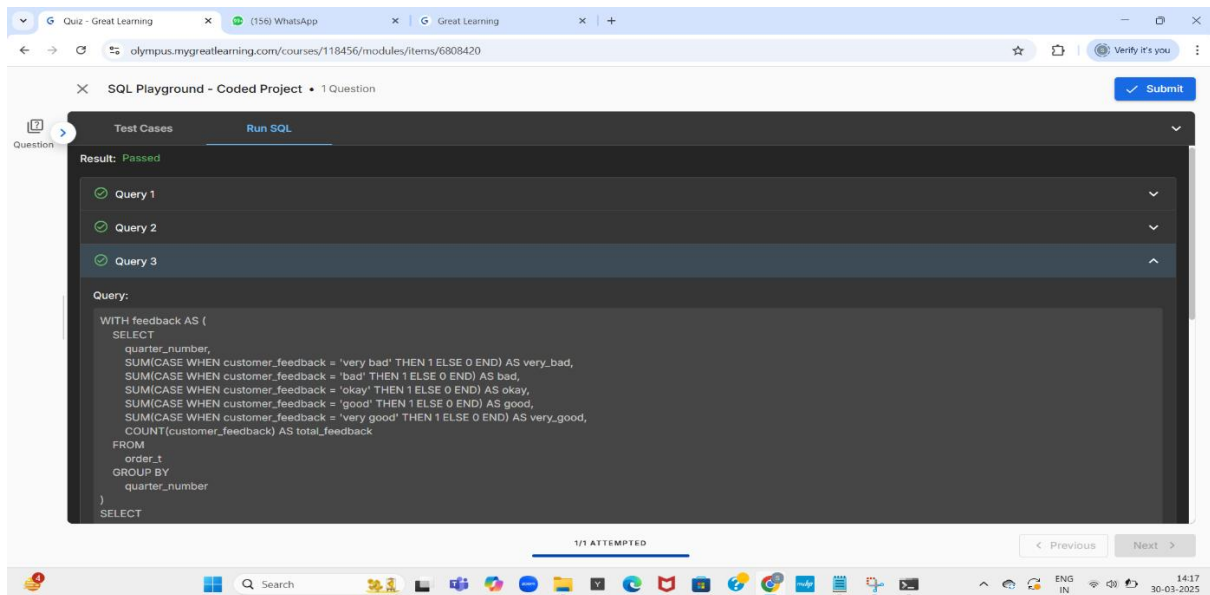
This indicates a **downward trend in customer satisfaction**.

Question 3: Which is the most preferred vehicle maker in each state?
[4 marks]

SOLUTION QUERY:

```
WITH feedback AS (  
  SELECT  
    quarter_number,  
    SUM(CASE WHEN customer_feedback = 'very bad' THEN 1 ELSE 0 END) AS very_bad,  
    SUM(CASE WHEN customer_feedback = 'bad' THEN 1 ELSE 0 END) AS bad,  
    SUM(CASE WHEN customer_feedback = 'okay' THEN 1 ELSE 0 END) AS okay,  
    SUM(CASE WHEN customer_feedback = 'good' THEN 1 ELSE 0 END) AS good,  
    SUM(CASE WHEN customer_feedback = 'very good' THEN 1 ELSE 0 END) AS very_good,  
    COUNT(customer_feedback) AS total_feedback  
  FROM  
    order_t  
  GROUP BY  
    quarter_number  
)
```

OUTPUT:



Observations and Insights:

The **percentage of "Very Bad" and "Bad" ratings is increasing**, while "Very Good" and "Good" ratings are decreasing:

- **Q1: 10.97% (Very Bad) → Q4: 30.65% (Very Bad)**
- **Q1: 30% (Very Good) → Q4: 10.05% (Very Good)**

This shows a clear **decline in customer experience**, potentially due to service issues, pricing, or product quality.

Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter? [5 marks]

SOLUTION QUERY:

```
SELECT
    quarter_number,
    ROUND(very_bad / total_feedback * 100, 2) AS Percentage_very_bad,
    ROUND(bad / total_feedback * 100, 2) AS Percentage_bad,
    ROUND(okay / total_feedback * 100, 2) AS Percentage_okay,
    ROUND(good / total_feedback * 100, 2) AS Percentage_good,
    ROUND(very_good / total_feedback * 100, 2) AS Percentage_very_good
FROM
    feedback
ORDER BY
    quarter_number;
select p.vehicle_maker, count(customer_id)
from product_t p
join order_t o on o.product_id = p.product_id
group by p.vehicle_maker
order by count(customer_id) desc
limit 5;
```

OUTPUT:

The top screenshot shows the 'Test Cases' tab in the SQL Playground. It lists four queries, all of which have passed. The bottom screenshot shows the 'Run SQL' tab, where the same query is executed, and the results are displayed in a table.

Query:

```
select p.vehicle_maker, count(customer_id)
from product_1 p
join order_1 o on o.product_id = p.product_id
group by p.vehicle_maker
order by count(customer_id) desc
limit 5
```

Output:

Showing 5 rows

vehicle_maker	count(customer_id)
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Observations and Insights:

Chevrolet (83 customers) is the most preferred, followed by:

- **Ford (63)**
- **Toyota (52)**
- **Pontiac (50)**
- **Dodge (50)**

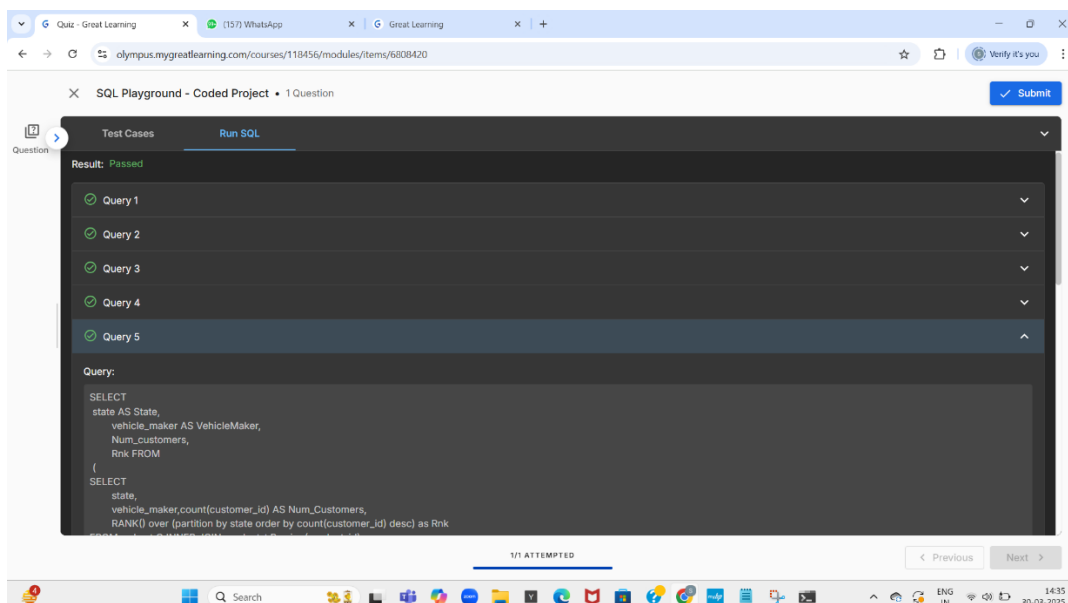
Chevrolet and Ford, being American brands, dominate customer preference.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time? [5 marks]

SOLUTION QUERY:

```
SELECT
state AS State,
    vehicle_maker AS VehicleMaker,
    Num_customers,
    Rnk FROM
(
SELECT
    state,
    vehicle_maker,count(customer_id) AS Num_Customers,
    RANK() over (partition by state order by count(customer_id) desc) as Rnk
FROM order_t O INNER JOIN product_t P using(product_id)
INNER JOIN customer_t C using(customer_id)
GROUP BY state,vehicle_maker
ORDER BY state) as Temp
WHERE Rnk =1;
```

OUTPUT:



The screenshots show a SQL Playground interface with a query and its results. The query is as follows:

```

Rnk FROM
(
SELECT
state,
vehicle_maker,count(customer_id) AS Num_Customers,
RANK() over (partition by state order by count(customer_id) desc) as Rnk
FROM order_t O INNER JOIN product_t P using(product_id)
INNER JOIN customer_t C using(customer_id)
GROUP BY state,vehicle_maker
ORDER BY state) as Temp
WHERE Rnk =1
  
```

The output shows the first 10 rows out of 143 rows:

State	VehicleMaker	Num_Customers	Rnk
Alabama	Dodge	5	1
Alaska	Chevrolet	2	1
Arizona	Pontiac	3	1
Arizona	Cadillac	3	1

The second screenshot shows the full result set of 143 rows:

State	VehicleMaker	Num_Customers	Rnk
Alabama	Dodge	5	1
Alaska	Chevrolet	2	1
Arizona	Pontiac	3	1
Arizona	Cadillac	3	1
Arkansas	Volkswagen	1	1
Arkansas	Suzuki	1	1
Arkansas	Pontiac	1	1
Arkansas	Mitsubishi	1	1
Arkansas	GMC	1	1
Arkansas	Chevrolet	1	1

Observations and Insights:

Chevrolet is most preferred in multiple states, including Texas, Ohio, Missouri, and others.

Other states have diverse preferences:

- **Toyota** is most preferred in **Florida, New York, and Oklahoma.**
- **Dodge** is popular in **Alabama, Idaho, and Mississippi.**
- **Mercedes-Benz** is preferred in **New Jersey, West Virginia, and Maine.**

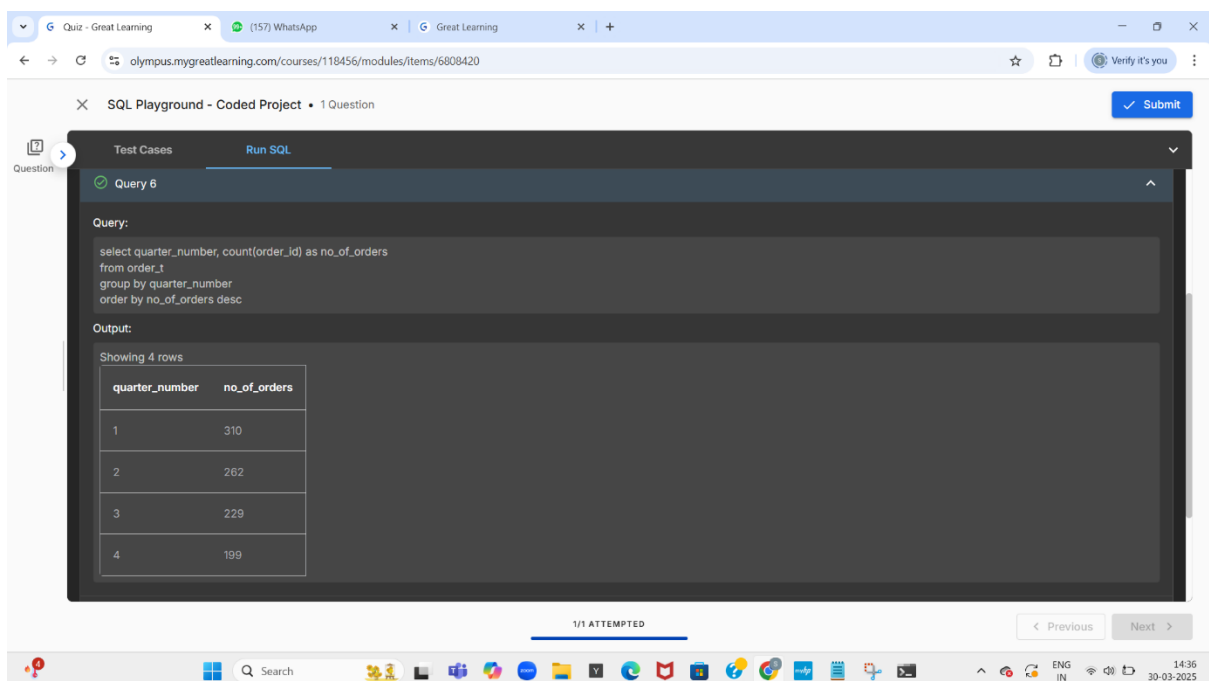
Preferences are somewhat linked to **regional brand availability and market segmentation.**

Question 6: What is the trend of the number of orders by quarter? [3 marks]

SOLUTION QUERY:

```
select quarter_number, count(order_id) as no_of_orders
from order_t
group by quarter_number
order by no_of_orders desc;
```

OUTPUT:



The screenshot shows a web browser window with the URL olympus.mygreatlearning.com/courses/118456/modules/items/5808420. The page is titled "SQL Playground - Coded Project" and contains a "Query 6" section. The query is:

```
select quarter_number, count(order_id) as no_of_orders
from order_t
group by quarter_number
order by no_of_orders desc;
```

The output is displayed as a table with 4 rows:

quarter_number	no_of_orders
1	310
2	262
3	229
4	199

The interface also shows a "Test Cases" tab, a "Run SQL" button, and a "Submit" button. The bottom of the screen shows a Windows taskbar with various application icons and the system clock indicating 14:36 on 30-03-2025.

Observations and Insights:

Orders are decreasing quarter-over-quarter:

- Q1: 310
- Q2: 262
- Q3: 229
- Q4: 199

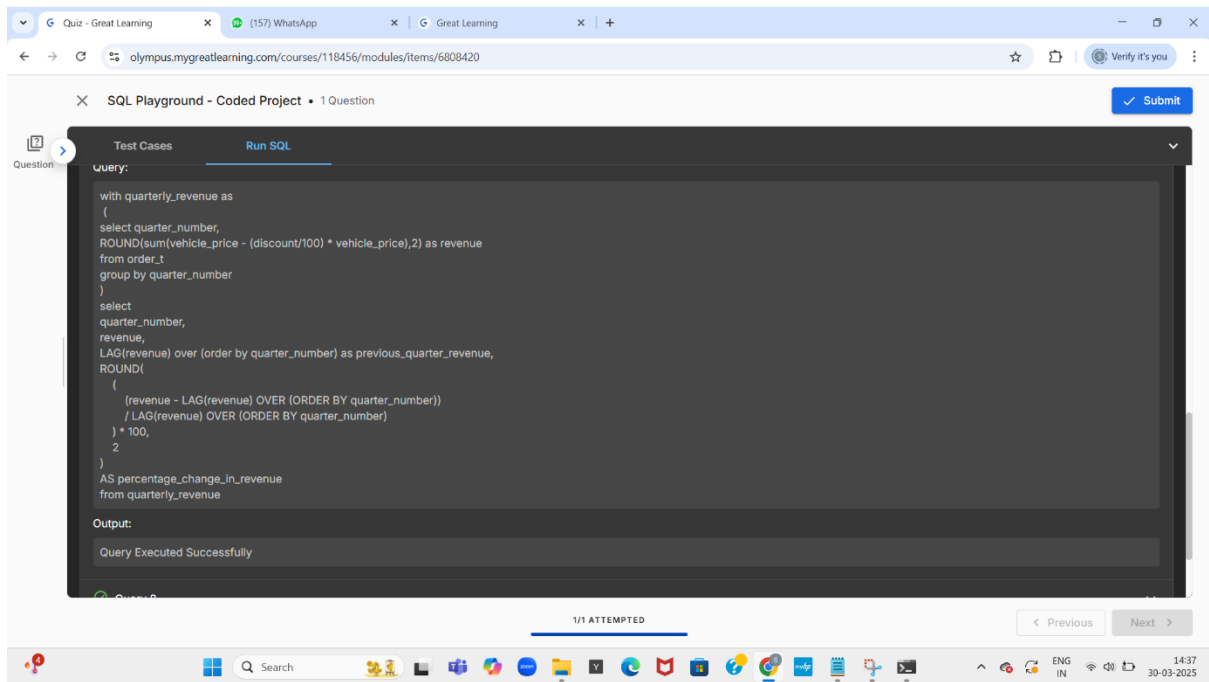
This trend aligns with **declining customer satisfaction and revenue**.

Question 7: Calculate the net revenue generated by the company.
What is the quarter-over-quarter % change in net revenue? [5 marks]

SOLUTION QUERY:

```
with quarterly_revenue as
(
select quarter_number,
ROUND(sum(vehicle_price - (discount/100) * vehicle_price),2) as revenue
from order_t
group by quarter_number
)
select
quarter_number,
revenue,
LAG(revenue) over (order by quarter_number) as previous_quarter_revenue,
ROUND(
(
(revenue - LAG(revenue) OVER (ORDER BY quarter_number))
/ LAG(revenue) OVER (ORDER BY quarter_number)
) * 100,
2
)
AS percentage_change_in_revenue
from quarterly_revenue ;
```

OUTPUT:



Observations and Insights:

Revenue is **continuously declining**, with the steepest drop in **Q4**:

- **Q1 → Q2: -17%**
- **Q2 → Q3: -10.65%**
- **Q3 → Q4: -20.13%**

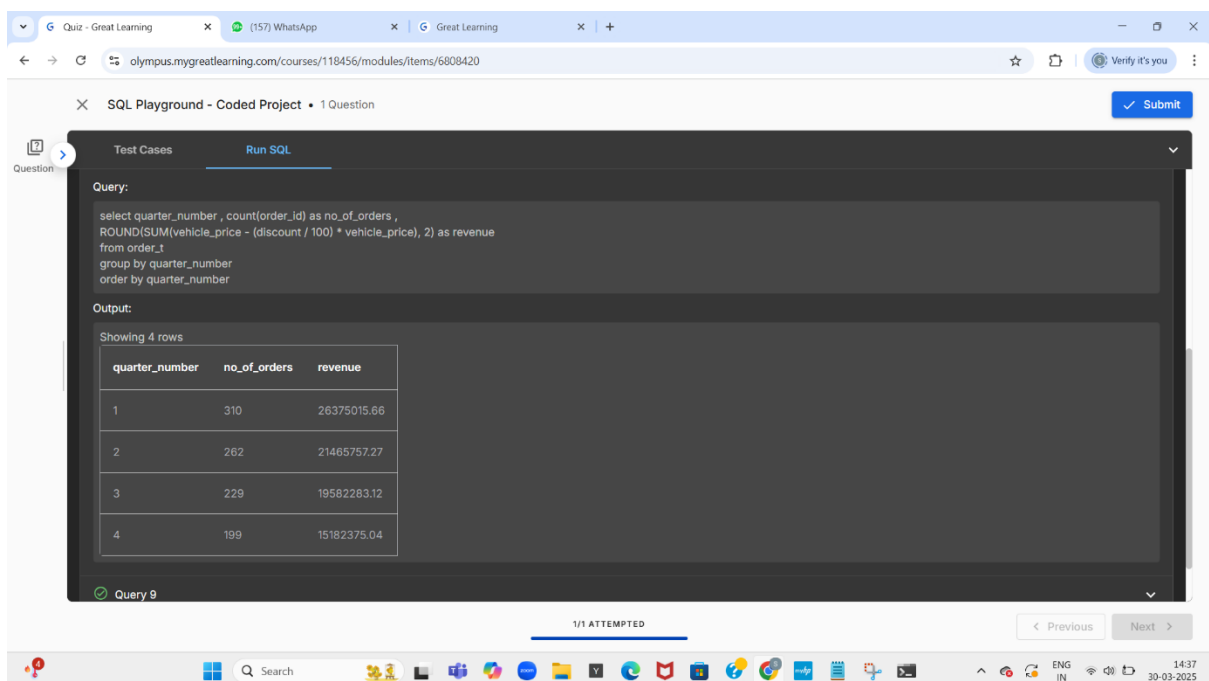
The declining revenue trend is concerning, indicating possible **customer churn, pricing issues, or competition**.

Question 8: What is the trend of net revenue and orders by quarters?
[4 marks]

SOLUTION QUERY:

```
select quarter_number , count(order_id) as no_of_orders ,  
ROUND(SUM(vehicle_price - (discount / 100) * vehicle_price), 2) as revenue  
from order_t  
group by quarter_number  
order by quarter_number;
```

OUTPUT:



The screenshot shows a web browser window with a SQL Playground interface. The query entered is:

```
select quarter_number , count(order_id) as no_of_orders ,  
ROUND(SUM(vehicle_price - (discount / 100) * vehicle_price), 2) as revenue  
from order_t  
group by quarter_number  
order by quarter_number
```

The output shows 4 rows of data:

quarter_number	no_of_orders	revenue
1	310	26375015.66
2	262	21465757.27
3	229	19582283.12
4	199	15182375.04

The interface also shows a 'Submit' button, a 'Test Cases' tab, and a 'Run SQL' button. The output is labeled 'Query 9' and '1/1 ATTEMPTED'.

Observations and Insights:

Revenue follows the **same declining trend as orders:**

- **Q1: \$39.42M → Q4: \$23.35M**

The correlation suggests that **fewer orders are directly impacting revenue**, rather than external factors like pricing changes.

Question 9: What is the average discount offered for different types of credit cards? [3 marks]

SOLUTION QUERY:

```
select c.credit_card_type,  
ROUND(  
avg(discount/100 *o.vehicle_price),2) as average_discount  
from order_t o  
join customer_t c on c.customer_id = o.customer_id  
group by c.credit_card_type  
order by c.credit_card_type;
```

OUTPUT:

Query 9

Query:

```
select c.credit_card_type,  
ROUND(  
avg(discount/100 *o.vehicle_price),2) as average_discount  
from order_t o  
join customer_t c on c.customer_id = o.customer_id  
group by c.credit_card_type  
order by c.credit_card_type
```

Output:

Showing first 10 rows out of 16 rows

credit_card_type	average_discount
americanexpress	517.37
bankcard	516.74
china-unionpay	523

1/1 ATTEMPTED

Previous Next

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credit_card_type	average_discount
americanexpress	517.37
bankcard	516.74
china-unionpay	523
diners-club-carte-blanc	503.24
diners-club-enroute	506.05
diners-club-international	491.22
diners-club-us-ca	500.54
instapayment	515.53
jcb	500.42
laser	557.24

Observations and Insights:

The highest discount is offered on **Laser cards (0.644)**, followed by:

- **Mastercard (0.63)**
- **Maestro (0.624)**

The lowest discounts are on **Diners Club International (0.584)** and **Solo (0.585)**.

This suggests a possible **credit card-based pricing strategy**.

Question 10: What is the average time taken to ship the placed orders for each quarter? [3 marks]

SOLUTION QUERY:

SELECT

'Q' || quarter_number AS Quarter,

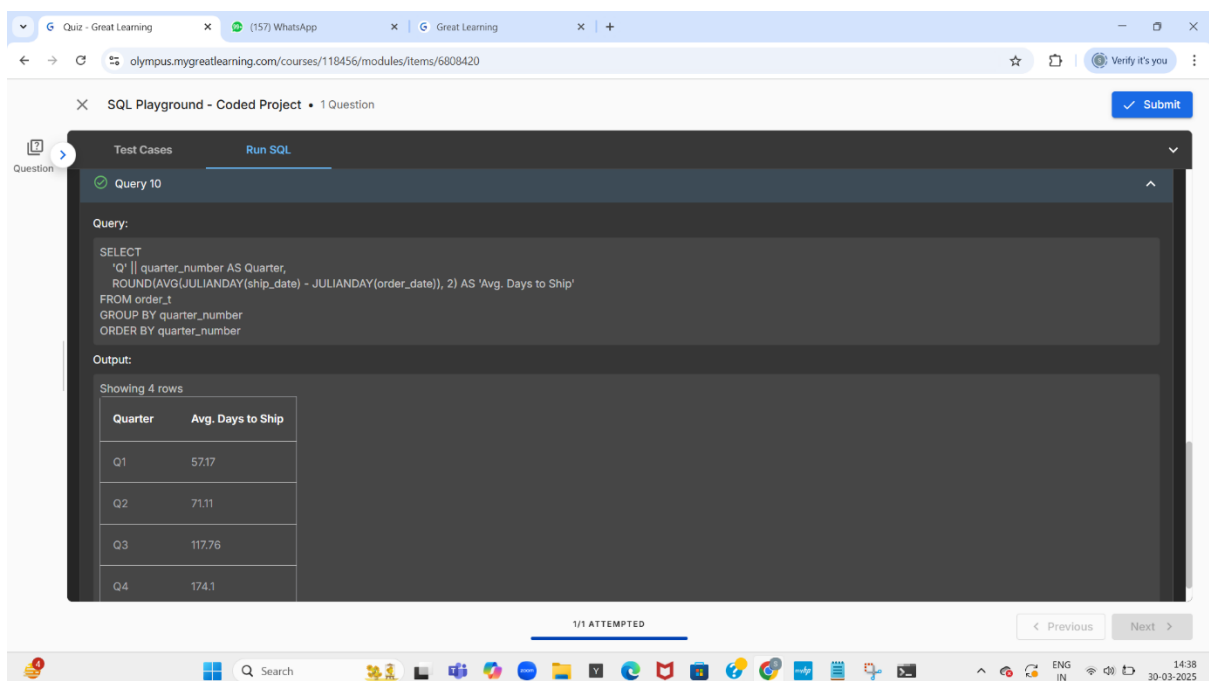
ROUND(AVG(JULIANDAY(ship_date) - JULIANDAY(order_date)), 2) AS 'Avg. Days to Ship'

FROM order_t

GROUP BY quarter_number

ORDER BY quarter_number;

OUTPUT:



The screenshot shows a web browser window with a SQL Playground interface. The query is displayed in a text area, and the output is shown in a table below it. The table has two columns: 'Quarter' and 'Avg. Days to Ship'. The data is as follows:

Quarter	Avg. Days to Ship
Q1	57.17
Q2	71.11
Q3	117.76
Q4	174.1

Observations and Insights:

Shipping time has drastically increased:

- **Q1: 57 days**
- **Q2: 71 days**
- **Q3: 118 days**
- **Q4: 174 days**

This significant delay is likely a **major contributor to declining customer satisfaction** and increasing negative ratings.

Business Overview

TOTAL REVENUE

124.71M

TOTAL ORDERS

1000

**TOTAL
CUSTOMERS**

994

AVG RATING

3.14

**LAST QTR.
REVENUE
2.33M**

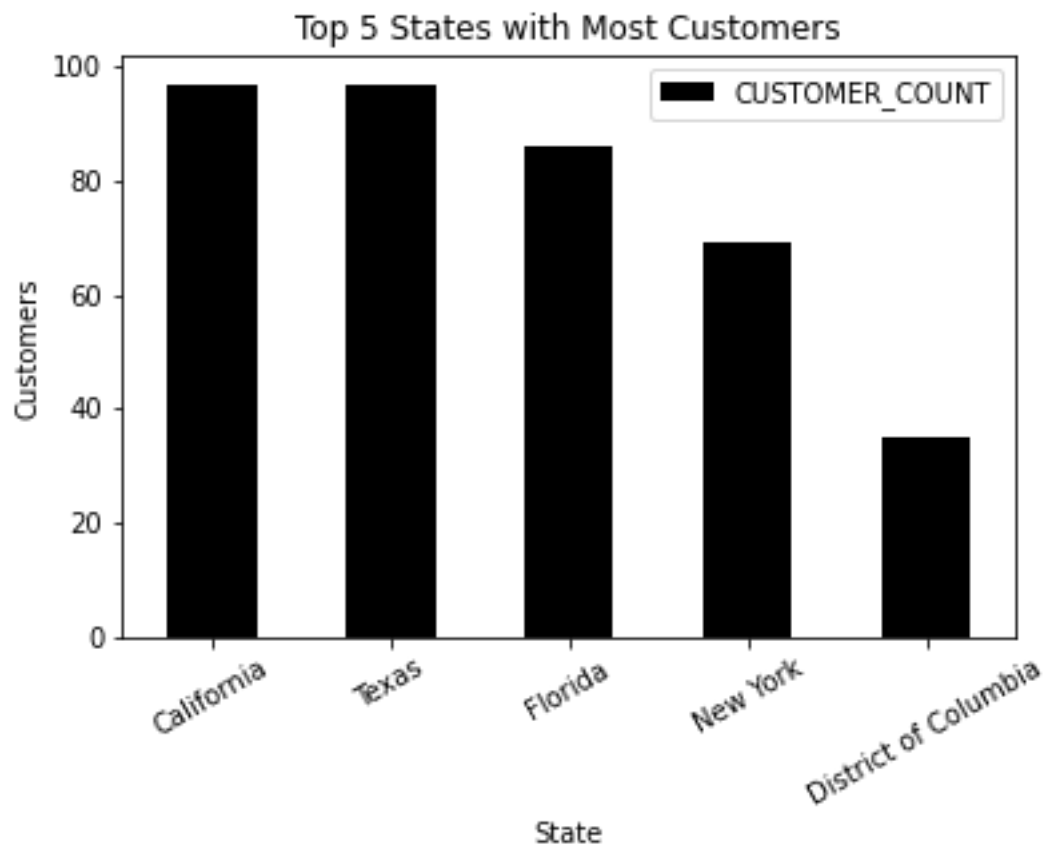
**AVG DAYS TO SHIP
98**

**LAST QTR.
ORDERS
199**

**% GOOD
FEEDBACK
44.1%**

CUSTOMER MATRICS

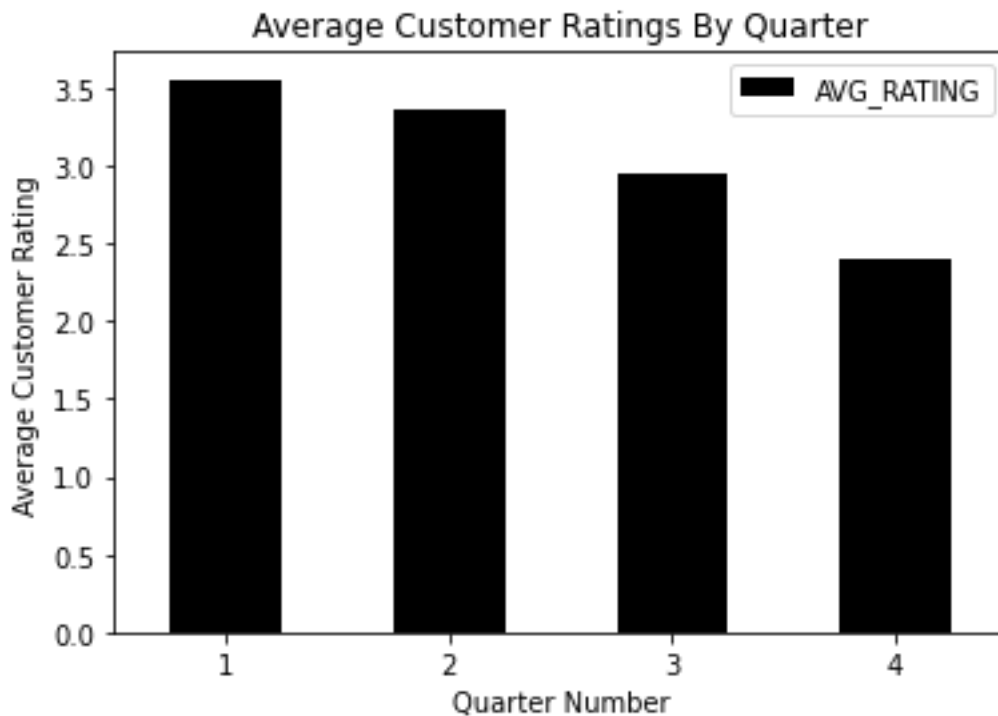
Distribution of Customers Across States



Observations:

- California and Texas have the most customers (97 customers)
- Followed by Florida (86), New York (69), District of Columbia (35)
- Maine, Wyoming, and Vermont have the least amount of customers (1 customer from each state)

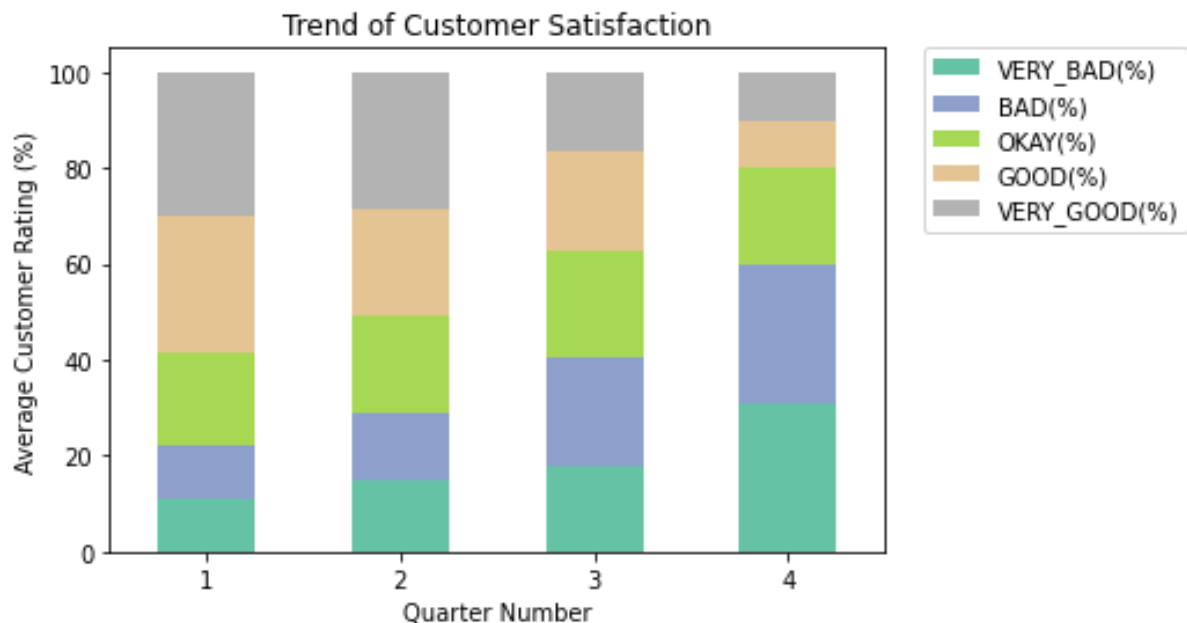
Average Customer Ratings by Quarter



Observations:

- Average ratings have decreased each quarter
- Average ratings are now at an all time low
- The biggest drop in average ratings was in between Quarter 2 and Quarter 3
 - Quarter 1: 3.55
 - Quarter 2: 3.35
 - Quarter 3: 2.96
 - Quarter 4: 2.40

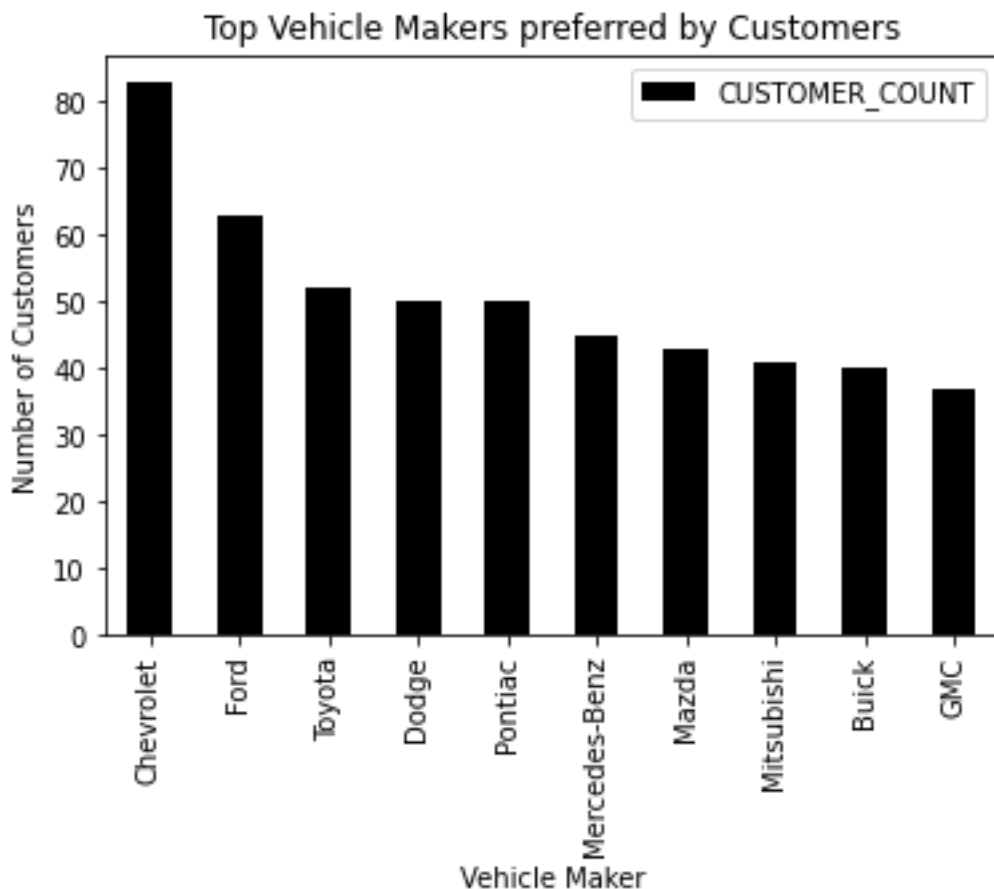
Trend of Customer Satisfaction



Observations:

- There is an increase in 'Very Bad' and 'Bad' ratings throughout the Quarters
- There is a decrease in 'Good' and 'Very_Good' Ratings throughout the Quarters
- Quarter 4 has the most 'Very Bad' and 'Bad' Ratings and the least amount of 'Good' and 'Very_Good' ratings compared to the other quarters

Top Vehicle Makers preferred by Customers

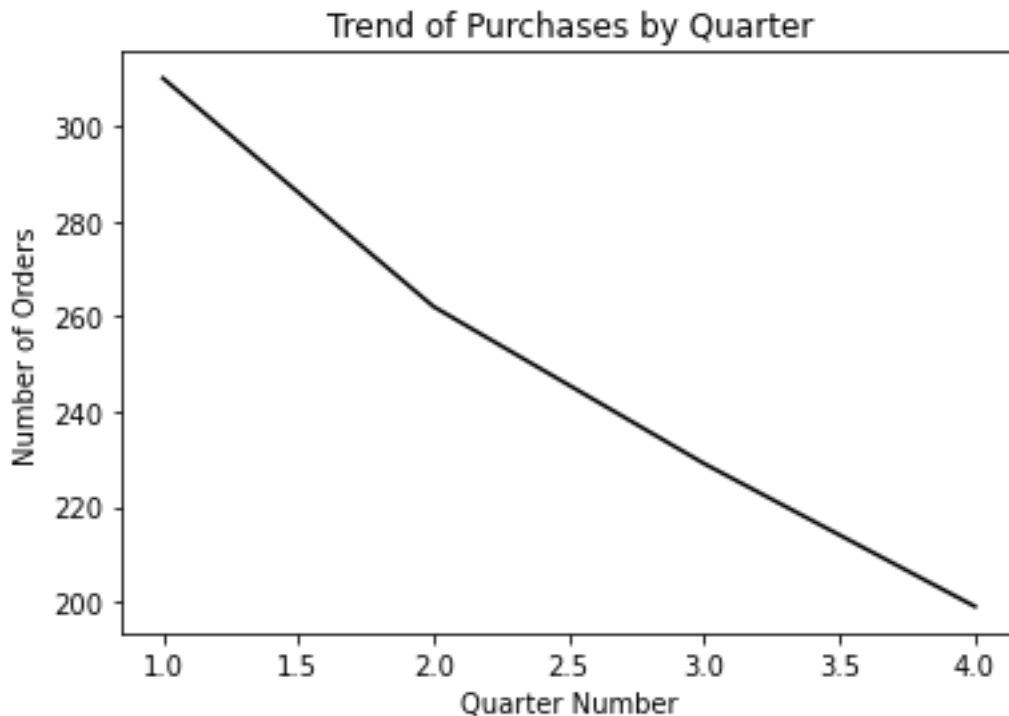


Observations:

- Chevrolet is the top favorite Vehicle Maker preferred by customers (83 customers)
- Ford (63 customers)
- Toyota (52 customers)
- Dodge and Pontiac (50 customers)
- Ram, MG, Daewoo are the least preferred vehicle makers by customers with only 1 customer each

Revenue Metrics

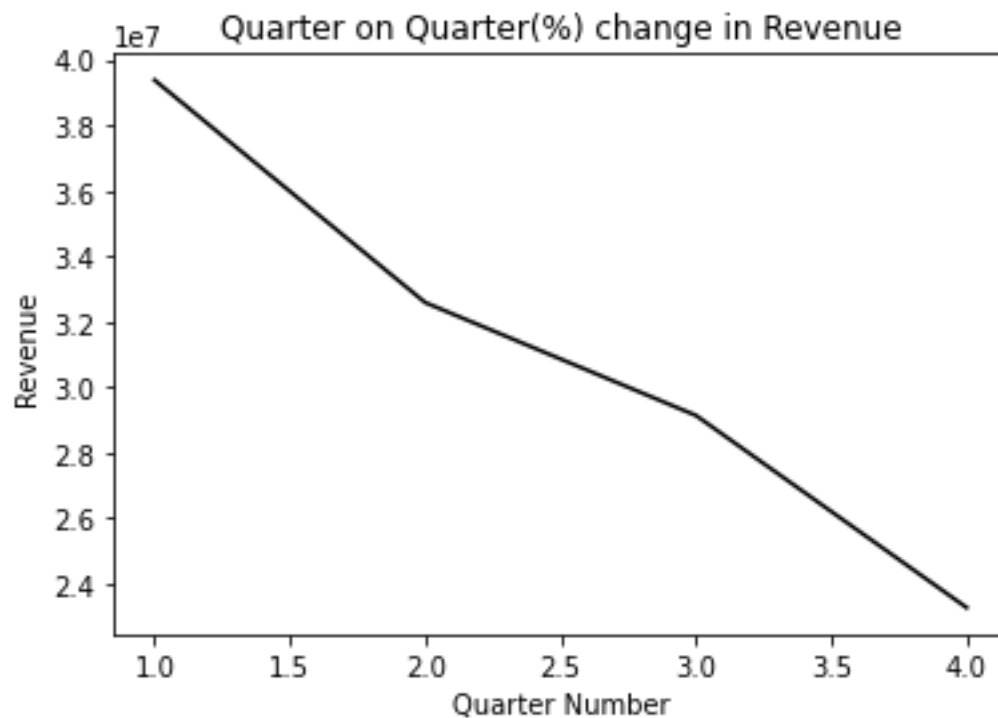
Trend of Purchases by Quarter



Observations:

- Over the quarters the number of orders have steadily decreased
- At Quarter 4 the company has reached its lowest number of orders
- More than 100 orders were lost between Quarter 1 and Quarter 4 (Now)
- The most orders lost were in between Quarter 1 and Quarter 2 (48 orders lost)

Quarter on Quarter % change in Revenue



Observations:

- Each quarter % change is negative meaning there was no increase in Revenue whatsoever
- The largest percentage decrease in Revenue was in Quarter 4
- The smallest percentage decrease in Revenue was in Quarter 3

Trend of Revenue and Orders by Quarter

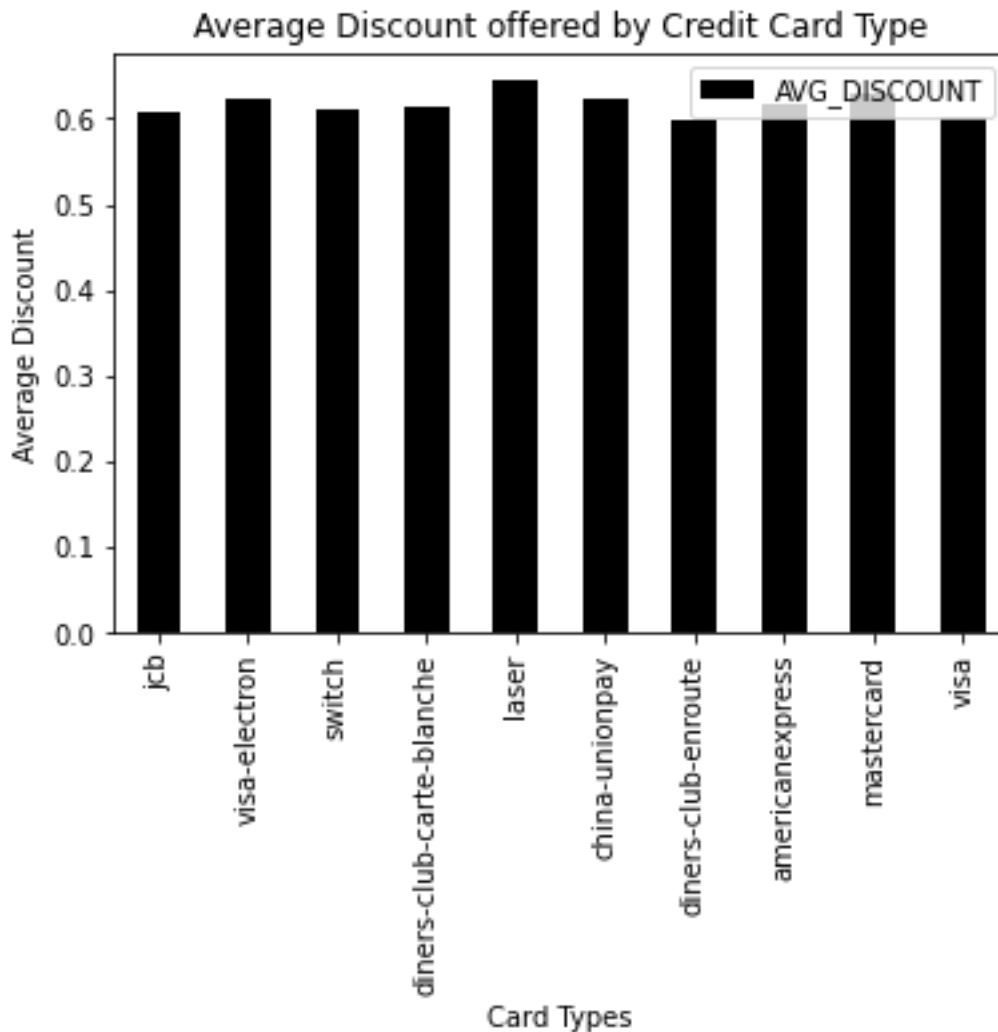


Observations:

- As orders decreased linearly by each Quarter, revenue also decreased linearly
- Quarter 1 was the peak of orders and the peak of revenue
 - Orders: 310
 - Revenue: \$39,374,179
- Quarter 4 is the lowest point for the company with the lowest amount of orders and lowest total revenue
 - Orders: 199
 - Revenue: \$23,276,651

Shipping Metrics

Average discount offered by Credit Card type

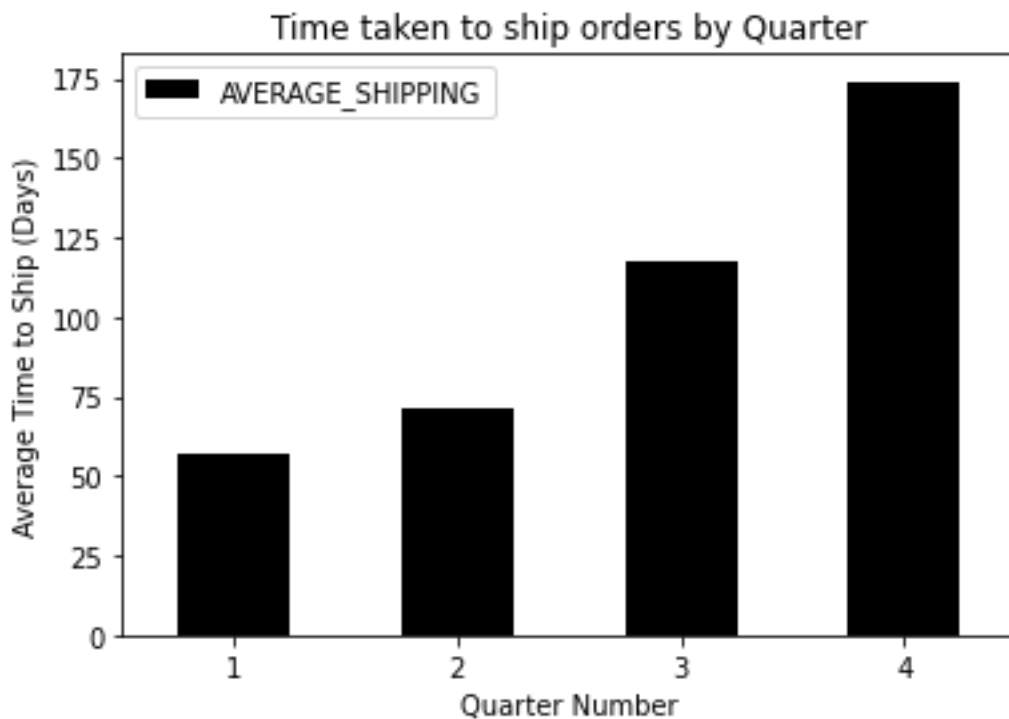


Observations:

- The credit card type with the highest average discount is Laser with a 64% discount
- The credit card type with the lowest average discount is Diners-club-international with a 58% discount

- The average discount offered by each credit card type ranges from 64% to 58%, with a majority of the discounts being over 60%

Time Taken to Ship Orders by Quarter



Observations:

- The average shipping time has gradually increased over each quarter
- Quarter 1 had the fastest shipping time averaging 57 days
- Quarter 4 had the slowest shipping time averaging 174 days
- The average shipping time has tripled since Quarter 1

Insight and Recommendations

Key Insights

1. Customer Satisfaction is Declining

- Average rating **dropped from 3.55 in Q1 to 2.40 in Q4.**
- Negative ratings (“Very Bad” & “Bad”) **increased from 10.97% to 30.65%.**
- This indicates **growing dissatisfaction, possibly due to delays or service issues.**

2. Orders & Revenue Are Falling

- Orders decreased by **35.8% from Q1 to Q4.**
- Revenue dropped by **40.8%, falling from \$39.42M (Q1) to \$23.35M (Q4).**
- The decline suggests **customer churn and decreased repeat purchases.**

3. Shipping Delays Have Tripled

- Average shipping time **increased from 57 days (Q1) to 174 days (Q4).**
- This is a major contributor to negative customer feedback.

4. Vehicle Preferences Vary by State

- **Chevrolet is the most preferred brand overall, but Toyota, Dodge, and Ford are strong regionally.**
- Chevrolet leads in **Texas & Ohio**, while Toyota is dominant in **New York & Florida.**
- **Luxury brands like Mercedes-Benz are preferred in New Jersey.**

5. Discount Strategy Differs by Payment Method

- **Laser, Mastercard, and Maestro cards get the highest discounts (~0.63).**
- **Diners Club & Solo have the lowest discounts (~0.58).**
- This suggests **pricing strategies may be influencing purchase behaviors.**

Key Recommendations

1. Improve Customer Satisfaction & Retention

- ✓ **Collect Customer Feedback:** Identify the main causes of dissatisfaction.
- ✓ **Enhance Customer Support:** Implement faster response times and proactive problem-solving.
- ✓ **Loyalty Programs:** Offer discounts or early access to returning customers.
- ✓ **Quality Assurance Checks:** Ensure product quality before shipping.

2. Boost Orders & Revenue

- ✓ **Re-engage Lost Customers:** Personalized emails and exclusive discounts.
- ✓ **Regional Targeted Ads:** Promote preferred vehicle brands in key states.
- ✓ **Seasonal Discounts & Flash Sales:** To drive demand in slow quarters.
- ✓ **Analyze Competitor Strategies:** Understand why customers are leaving.

3. Fix Shipping Delays (Biggest Issue)

- ✓ **Partner with Reliable Logistics Providers:** Reduce lead times.
- ✓ **Offer Expedited Shipping Options:** Customers willing to pay extra for faster delivery.
- ✓ **Improve Supply Chain Planning:** Keep high-demand vehicles in stock.
- ✓ **Real-Time Tracking & Proactive Updates:** Inform customers before they complain.

4. Optimize Vehicle Inventory & Marketing

- ✓ **Stock Chevrolet & Ford models more in high-preference states.**
- ✓ **Expand Toyota & Luxury Offerings in states where they are preferred.**
- ✓ **Use Data to Personalize Offers:** Promote the right cars in the right regions.

5. Refine Discount & Payment Strategies

- ✓ **Analyze High-Discount Cards:** Are Laser/Mastercard users more profitable? If not, adjust pricing.
- ✓ **Promote Payment Methods That Drive Sales:** Offer better incentives on profitable card types.

Final Takeaway

- **Urgent Fix: Improve Shipping Speeds & Customer Experience** (Biggest reason for declining sales).
- **Short-Term Priority: Re-engage lost customers with discounts & better service.**
- **Long-Term Growth: Use data-driven marketing & inventory planning for better brand positioning.**

THANK YOU

HIYA SHAH